

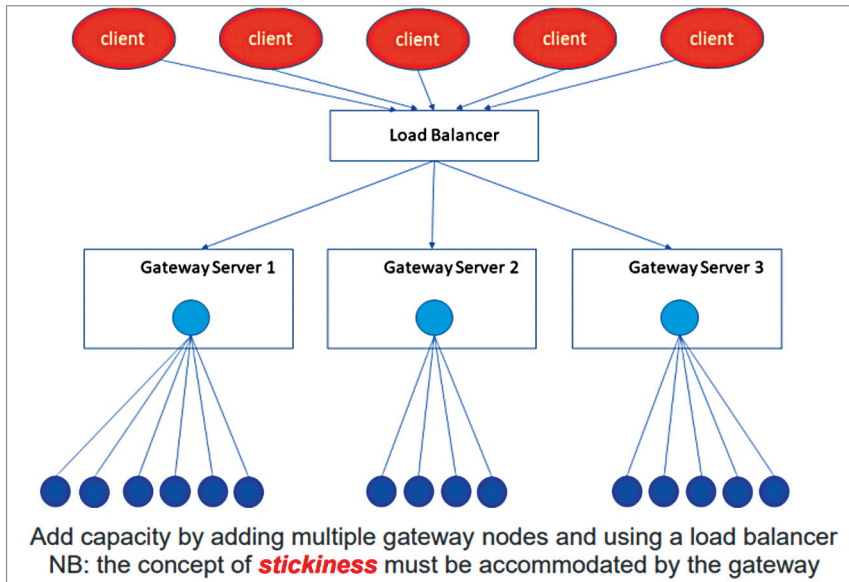


Fig. 17: Capacity building by adding gateway nodes

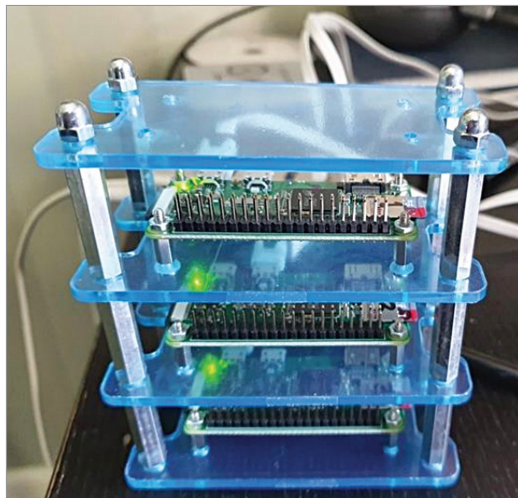


Fig. 18: Rack-mounted Raspberry Pi Zero W multi-node solution made by the author

concurrent connections that a central device can establish with a peripheral device, there is always a limit in practice, and it can be quite low.

So here are some options if you want to support a large number of devices that are concurrently connected to and controlled by the Bluetooth Internet Gateway.

1. Adding multiple adapters. You can add multiple Bluetooth adapters (Fig. 16) to your gateway. Some adapters come in the form of USB dongles, so physically that will be easy to implement; some tweaks to your code will be needed but that wouldn't be hard.

2. Horizontal scaling. This was the approach taken in this project.

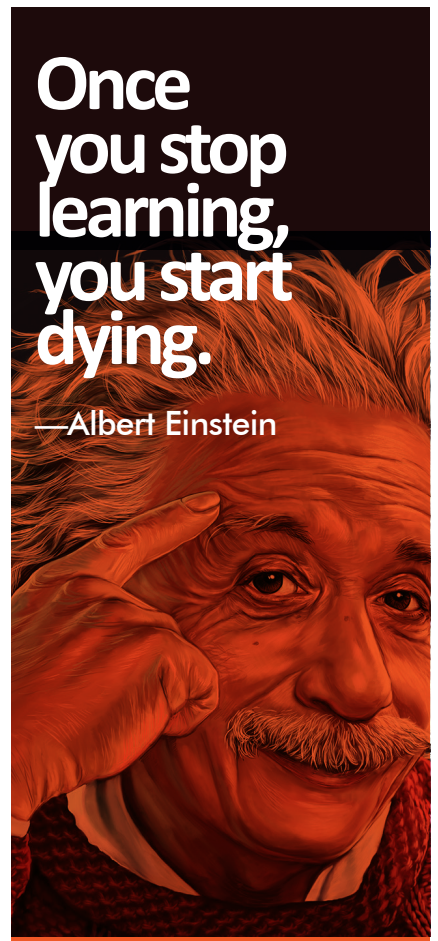
Multiple instances of gateway services (Fig. 17) were run on multiple Raspberry Pi devices and requests distributed across them by a software load balancer. All client requests went to the node running the load balancer.

Using horizontal scaling, in theory, a large number of Raspberry Pi nodes could have been added to the gateway cluster and very-high concurrent Bluetooth device support achieved. The only issue to be aware of here was that there is a load balancer concept known as stickiness. Once a Bluetooth connection has been established from

one of the gateway nodes to a particular Bluetooth device, all subsequent requests from the clients that relate to that Bluetooth device must be handled by the same gateway node since it holds the Bluetooth connection.

The HAProxy load balancer has excellent support for stickiness, allowing one to configure the behaviour. **EFY**

This article is based on the talk Understanding reliability in Bluetooth technology by Martin Woolley, Senior Developer Relations Manager, EMEA at Bluetooth SIG during February edition of Tech World Congress and India Electronics Week 2021



Once you stop learning, you start dying.

—Albert Einstein

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